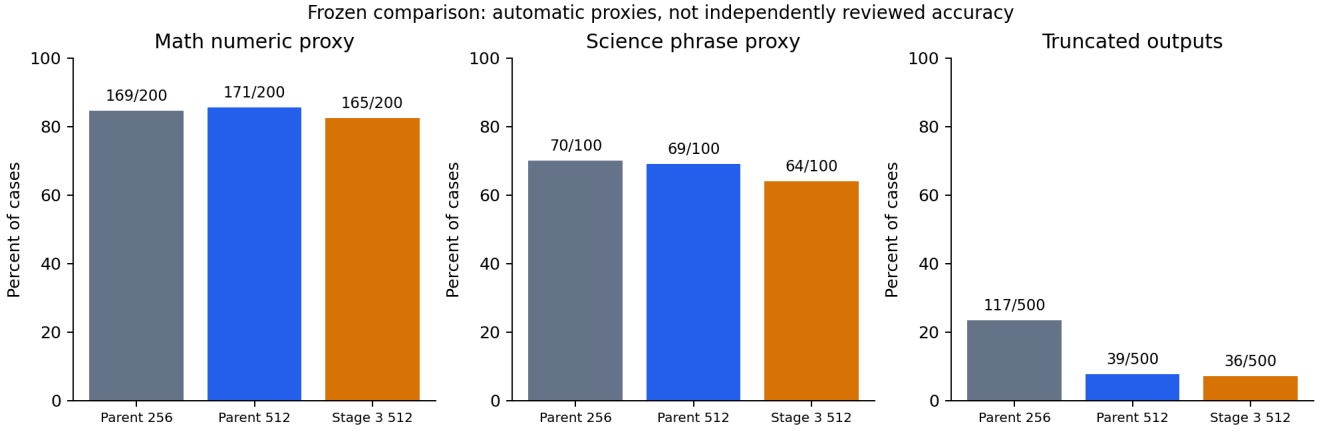
MoE Routing and Adaptation

Research and engineering report · October 22, 2026

Research release. Experiments complete and frozen. Stage 3 not promoted. Answer-quality and production acceptance false; independent human review pending.

A study of request-level routing, native expert/gate adaptation, checkpoint recovery, numerical export checks and authenticated serving.



Aggregate evidence and original artifact identities accompany this report in the source release. A successful export or operating check is distinct from answer quality.

Verified results and release scope

Research frozen October 22, 2026. This is a research and engineering project; answer-quality and production acceptance remain false. Independent human review is pending. This portable summary uses aggregate results; the original detailed report and machine evidence remain preserved in the research workspace.

Systems studied

Stage 1 uses an external DeBERTa request router and named domain/safety LoRA adapters on a shared Llama-3.1-8B-Instruct base. Seventeen external weight files and the original source ZIP passed preservation checks. The historical 98.04% router score came from a split with shared questions and is not an independent generalization result.

Stage 2 adapts GPT-OSS-20B attention projections while freezing native FFN experts and gates. Training completed 1,489 optimizer steps using 23,814 eligible examples; checkpoint 300 was selected and the attention update scaled to 0.125.

Stage 3 tests updates to actual native FFN experts, with and without gate updates. Both pilots used seed 42, rank 2/alpha 2, effective batch 2, maximum length 512, expert LR 1e-5 and 200 steps. Joint gates used LR 1e-6 and warmup 20. The frozen development pool had 1,000 training and 80 validation rows from five sources. These short pilots do not establish multi-seed robustness or full convergence.

Development and export results

Parent development NLL was 2.581872655. Unmerged expert-only and joint-gate pilots reached 1.745005778 and 1.828360192 respectively. Both original full-scale exports failed the unchanged numerical gate.

Calibrated export	Scale	Merged development NLL	Numerical qualification
Expert-only	0.5	2.112657	Failed merge KL
Expert-only	0.25	2.407196	Failed cached/full KL
Joint gates	0.5	2.187788	Passed
Joint gates	0.25	2.427826	Passed

The fixed development-only selection rule chose joint gates at 0.5, a 15.3% development NLL reduction. The standard archive was independently restored with 22 files verified. Five numerical probes and exact reload checks support this export result; they do not qualify an arbitrary serving engine or overall quality.

Frozen 500-prompt comparison

Each arm generated 500 unique, nonempty, error-free answers. All 500 input-token hashes matched between precision-parent and Stage 3. Greedy decoding, seed 42, fixed template date, a 2,048-token context envelope and the preserved HF eager BF16 environment were used. Baseline-to-precision changes both instructions and output budget; it does not isolate either factor.

Arm	Math numeric proxy /200	Science phrase proxy /100	Truncated /500
Stage 2 parent, baseline 256	169	70	117

Arm	Math numeric proxy /200	Science phrase proxy /100	Truncated /500
Stage 2 parent, precision 512	171	69	39
Stage 3 selected, precision 512	165	64	36

Against the matched parent, Stage 3 has nine math gains and fifteen losses, for a net loss of six proxy matches; science loses five matches. Three fewer outputs truncate. Numeric/phrase matching is an automatic proxy, not independently assessed accuracy. No held-out proxy improvement was demonstrated. The candidate was not retuned after these results and is not promoted.

The suite contains 200 math, 100 science, 100 general, 40 support, 40 publisher- harmful and 20 publisher-benign-proxy prompts. Exact normalized project overlap checks passed; semantic overlap and foundation pretraining exposure are unknown. Publisher categories are not independently adjudicated safety denominators.

Operating evidence and quality limits

Stage 2 job 6403909 previously verified archive cold start in 452.7 seconds, 10,072 requests over two hours, p95 completion latency 2.898 seconds, native restart in 299.7 seconds and external rollback in 198.6 seconds. These are workload-specific operating measurements, not concurrent-user capacity claims. The recorded workload used concurrency four, 512 input tokens and a 128-token output budget; these timings must not be extrapolated to arbitrary workloads. That service was stopped; these measurements do not establish a live deployment or qualify the Stage 3 profile.

Earlier Stage 2 assessment covered 526 answers with attributed unblinded AI assessment, including exact assessment reuse. General 49/100 and support 15/40 met the 4/5 threshold, below required 80/100 and 32/40. Benign-sensitive refusals were 7/20 against a maximum of one. Those failures remain open. A later 54-answer AI qualitative review found concrete errors and overrefusals; it supplies examples, not full-suite or human-reviewed rates. No clinical certification is claimed.

Provenance anchors

These SHA256 identifiers bind the original research evidence. The upcoming public release must include selected source/evidence exports so that these anchors do not substitute for accessible reproduction materials.

Artifact	SHA256
Frozen comparison protocol	`7a06ef743e0e7c2aa13ded019c3ffe4a9f1ee9358cae22030fd9fc00020b0f59`
Frozen prompt suite	`89ae51c086b4f70b56ad9d6ab361a06ce2a6d7d79368226ad8caf5c47a7a5da9`
Parent baseline answers	`ad2bd8566223c4d9adba6549dc1bbf79059a96bd7487395cac75d386425eb3a9`
Parent precision answers	`965460a7c18e23a38e6de621c1aa49fb8fe6e8a8e5ee49c37d4ef6e005f28556`
Stage 3 answers	`46342f9ea5e61cded0f19aa8b037717054a523340239979678db392f47f2e781`
Selected Stage 3 archive	`4b8703a040e54455597ad1e218b5e2a846cb04d930be35d8d37d8ee0cb38f8e5`

Collection jobs: calibration 6447695, parent 6447696, candidate recovery 6477680, final verification/report 6477681; all completed under normal coc-ice. Initial candidate job 6447697 failed CUDA access before generation; failed evidence was retained and the unchanged candidate recovered without repeating parent arms.

The engineering contribution is the implemented and verified adaptation, checkpoint/resume, export, evaluation and operating workflow. Lower development loss did not translate into a held-out proxy advantage. The public release will present that result and its limitations explicitly.

Technical case study: adaptation that did not earn promotion

Question

Can a working native MoE service be extended with real FFN-expert and gate updates while preserving checkpoint recovery and portable deployment, and does the resulting candidate improve held-out answers over its parent?

The implementation question and the quality question have different outcomes. Real expert/gate training, resumability and a calibrated standard export worked. Held-out automatic math/science proxies did not improve.

Engineering decisions

The external system uses one shared base with explicitly named adapters and serialized adapter selection. This avoids a mutable-model concurrency bug in which the reported specialist and actual adapter can diverge. Each checkpoint's label mapping is validated rather than assumed.

Native serving places a bounded authenticated gateway before a separate private inference engine. Engine authentication covers health, metrics and other paths; gateway requests have input/output envelopes, queue limits and cancellation. Operational probes check disconnect cleanup and profile-scoped history.

Before expensive pilots, zero-update parity and split-run optimizer, scheduler and RNG resume equivalence were tested. Best/latest checkpoints retain training state. GPU work uses a single suitable accelerator; immutable models are staged on inspected node-local storage, with persistent artifacts kept in scratch.

Controlled experiment

Two 200-step pilots used the same development pool and seed: expert-only versus expert-plus-gate updates. Full-scale exports failed the original numerical KL threshold even though unmerged development losses improved. Rather than relaxing the threshold, a fixed four-trial scale calibration selected the lowest-loss export that passed all original numerical/reload/cache checks.

The selected joint-gate scale-0.5 export was independently restored. Selection finished before a reserved 500-prompt suite was collected. Parent and candidate precision arms have identical input-token hashes on every case. The baseline arm changes both prompt instructions and output budget and is analyzed separately.

Outcome and decision

Development NLL improved 15.3%, but the selected candidate had six fewer math numeric matches and five fewer science phrase matches than its matched parent. Truncation improved by only three outputs. An attributed AI review found concrete regressions, including invented inventory in an arithmetic answer, alongside some better contextual handling. This is error analysis, not independent scoring.

The decision was to retain the artifact for research and keep the qualified parent as the initial demo target. The parent itself still has answer-quality failures, so operating qualification is not described as production acceptance.

Recovery and interpretation

An initial candidate allocation failed CUDA access before generation. A short earlier diagnostic had been insufficient evidence to clear the problematic node exclusion permanently. Recovery restored the exclusion and reran only the unchanged candidate, preserving parent answers and failed logs. Root cause was not established as a node-wide hardware fault.

This study demonstrates a working experimental and operating workflow. It does not establish that gate training is better than frozen gates, that the candidate is safe, or that native expert adaptation cannot work. One short seed and five numerical probes leave substantial uncertainty. Further model tuning is stopped for this release; the measured negative result is the conclusion.

See results, evaluation methods and the system card.

Evaluation methods and independent review

Frozen comparison

The suite has 200 math, 100 science, 100 general, 40 support, 40 publisher-harmful and 20 publisher-benign-proxy prompts. The protocol, prompt suite and three raw answer hashes are recorded in `../reports/results.json` and the results report. All 1,500 outputs passed identity, uniqueness, nonempty/error and configuration checks. All 500 paired precision input hashes agree.

Settings: seed 42, greedy decoding, fixed template date 2026-10-19, 2,048-token context envelope, HF eager BF16, H100 model class. The precision arms use a 512-token collection budget. The baseline uses 256 and different instructions. The frozen offline comparison budget is distinct from the gateway's 256-token serving envelope; API demonstrations are not reproductions of those 512-token outputs. No candidate was tuned or reselected after held-out answers were read.

Math scoring is a numeric-answer proxy and science scoring is a phrase-match proxy. Consult the preserved evaluation source for exact extraction rules. Truncation reflects the declared generation budget and is not a correctness judgment. The report makes no statistical superiority claim.

Existing review

Earlier Stage 2 quality assessment used attributed, unblinded AI review, including exact reused assessments; it failed required gates. Later qualitative review explicitly read 18 cases across three arms (54 answers), selected deterministically after collection. It is illustrative AI error analysis, not a preregistered scoring study or an independent human evaluation.

Independent review protocol

The private workspace contains a masked 1,500-answer CSV and separate condition crosswalk. A reviewer must not inspect that crosswalk while scoring. Full-suite review is preferred. If resources allow only a subset, freeze a stratified sample and its scope before scoring, and report subset results without full-suite claims.

Record reviewer identity or attributable reviewer ID, review date, rubric version, and whether they contributed to model development. Score correctness, instruction following, completeness and relevance separately. Preserve uncertain/unassessable cases instead of forcing passes. For support, assess contextual relevance and unsupported claims without claiming clinical validity.

Adjudicate prompt intent separately from answer behavior: distinguish harmful facilitation, legitimate educational inquiry, reported prejudice and distress. Publisher labels alone cannot establish benign overrefusal or harmful-compliance denominators. Record disagreements and adjudication with an audit trail. Where multiple reviewers are available, report agreement before consensus.

No scores have been supplied by an independent human reviewer. Research publication may state this limitation; it must not invent completion or waive known quality failures. No new training is part of this review.

System card

Intended use

Research demonstration and examination of routing, adaptation, export, evaluation and inference engineering. The source is suitable for studying the implementation. The models are not accepted as general production assistants, clinical systems, safety classifiers or reliable sources of current advice.

Profiles

Profile	Base and updates	Release role
External Stage 1	Llama-3.1-8B-Instruct, DeBERTa router, named LoRA specialists	Preserved research profile and rollback
Native Stage 2	GPT-OSS-20B, calibrated attention update	Initial restricted GPU demo target; prior operations passed
Native Stage 3	Stage 2 plus calibrated FFN/gate updates	Research artifact; not promoted or serving-qualified
CPU demo	Deterministic service double	UI/API demonstration only; no model inference

Data and evaluation

Training/evaluation sources include GSM8K, SciQ, Dolly, PKU and ESConv, with additional external-router/safety sources recorded in preparation scripts. Dataset provenance and allowed use are separate from the code license. Raw conversations, held-out prompts/answers and weights are not included by default.

The held-out comparison is 500 prompts across six categories and three arms. Exact project-overlap checks passed; semantic overlap and pretraining exposure are unknown. Automatic numeric/phrase proxies have known false positives and false negatives. Independent human scoring and safety-intent adjudication are pending. No clinical or independent human certification is claimed.

Known failures

The Stage 2 assessment failed General and support quality thresholds and the benign-sensitive overrefusal limit. Stage 3 did not demonstrate a held-out proxy gain and introduced specific arithmetic regressions. Support outputs frequently truncate; generated claims and regional resources may be unsupported. Source labels do not automatically define the intended harmful/benign denominator. The historical external router score had train/test question overlap.

Access and operation

Real demos require authenticated access on a loopback listener through an SSH tunnel. Keep engine credentials separate from the application credential. Tokens, request content and access-setting fields must not appear in recordings or public logs. GPU inference belongs on an allocated compute node.

The server has bounded inputs, a queue/concurrency envelope, cancellation and profile-scoped history. These are operating controls, not proofs of safe model answers. ICE allocations expire; documented historical measurements are not evidence that a service is live. See the deployment status record.

Release decision

Research release only. Production accepted: **false**. Answer quality accepted: **false**. Independent human review: **pending**. Changes to these states require new attributable evidence, not a new README or successful scheduler exit.